



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2017 Preliminary Course Yearly Assessment Task
Wednesday, 30 August 2017

General instructions

- Working time – **150 minutes**.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt all questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided

SECTION II

- Commence each new question on a **new page**. Write on both sides of the paper.
- **All necessary working should be shown** in every question. Marks may be deducted for **illegible or incomplete working**.

STUDENT NUMBER: **# BOOKLETS USED:**

Class: (please ✓)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa

- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris

Marker's use only

QUESTION	1-5	6	7	8	9	10	11	12	Total
MARKS	5	15	11	16	12	18	9	10	96

Section I

5 marks

Attempt Question 1 to 5

Mark your answers on the answer grid provided

1. What is the number of asymptotes to the graph of

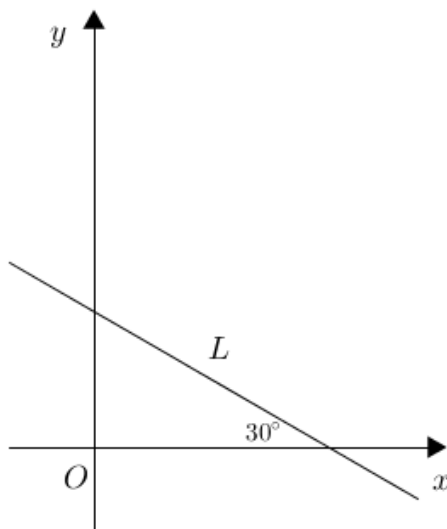
$$y = \frac{x^2}{x^2 - 4x - 12}$$

- (A) 1 (B) 2 (C) 3 (D) 4

2. What is the focus of the parabola $(x + 3)^2 = 12(y - 3)$?

- (A) $(-3, 3)$ (B) $(0, 3)$
(C) $(-3, 6)$ (D) $(-3, 0)$

3. The diagram shows the line L .



What is the slope of the line L ?

- (A) $-\sqrt{3}$ (B) $-\frac{1}{\sqrt{3}}$
(C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

4. What is the value of $\tan(\alpha - \beta)$ if $\sin \alpha = -\frac{3}{5}$ for $\frac{3\pi}{2} < \alpha < 2\pi$ and $\sin \beta = \frac{4}{5}$ for $\frac{\pi}{2} < \beta < \pi$?

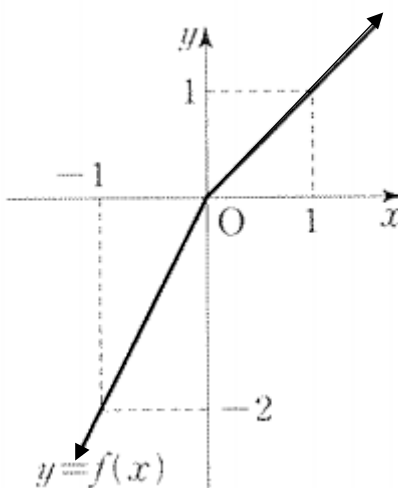
(A) $-\frac{4}{3}$

(B) $-\frac{5}{24}$

(C) 5

(D) $\frac{7}{24}$

5. The graph of the function $y = f(x)$ consists of two rays that pass through the origin and the two points $(1, 1)$ and $(-1, -2)$ as shown below. Choose the **TRUE** statements from the following.



Statements

a) $f^{-1}(5) = f^{-1}(f^{-1}(5))$

b) $f^{-1}(-2) = -1$

c) There is only one solution for $f^{-1}(x) = f(x)$

(A) a, b

(B) a, b, c

(C) a, c

(D) b, c

End of Section I.

Examination continues overleaf.

Section II

91 marks

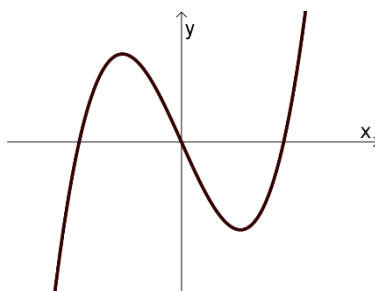
Attempt Question 6 to 12

Write your answers in the writing booklets supplied. Additional writing booklets are available.
Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 marks)

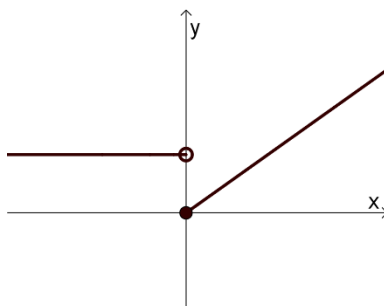
- (a) Copy the following graphs, then sketch their gradient function on a set of axes directly below each graph:

(i)



2

(ii)



2

- (b) Differentiate $f(x) = 3x^2 - 7x$ from first principles.

3

- (c) Differentiate and **FULLY** simplify the following

(i) $y = 4x^2 + \frac{3}{\sqrt{x}}$

2

(ii) $y = (1 - 2x)^5 \cdot (2x + 1)$

3

(iii) $y = \frac{5x^2 + 3}{1 - x}$

3

Question 7 (11 marks)

- (a) Find the equation of the normal to the curve $y = 2x^3 - 10x + 1$ at $x = 2$. 3
- (b) For the curve $y = -x^3 - x^2 + 16x - 20$,
- (i) Find and classify any stationary points. 3
- (ii) Find any points of inflexion. 3
- (iii) Sketch the graph of $y = -x^3 - x^2 + 16x - 20$. (You are not required to find the x-intercepts) 2

Question 8 (16 marks)

- (a) If α and β are the roots of $3x^2 - 4x + 2 = 0$, without solving the equation find the values of:
- (i) $\alpha + \beta$ 1
- (ii) $\alpha\beta$ 1
- (iii) $\frac{2}{\alpha} + \frac{2}{\beta}$ 2
- (iv) $(\alpha - \beta)(\beta - \alpha)$ 2
- (b) If $3x^2 - 2x + 5 \equiv Ax(x - 2) + B(x - 2) + C$, find A , B and C . 3
- (c) Solve $3^{2x} - 12(3^x) + 27 = 0$, giving all answers in exact form. 2
- (d) For what values of m are the roots of $x^2 + 4x - 3 = m(x - 2)$
- (i) real? 2
- (ii) both negative? 3

Question 9 (12 marks)

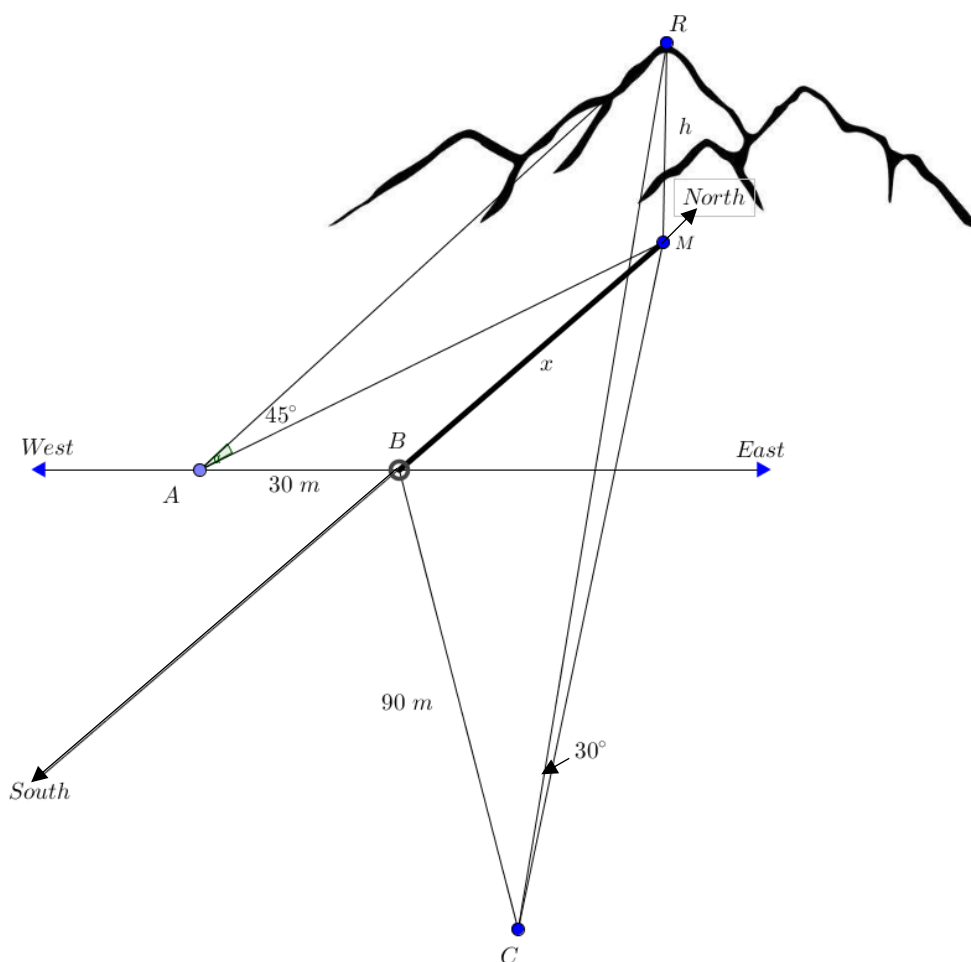
(a) Find the exact value of $\operatorname{cosec} 240^\circ$ 1

(b) Sketch the curve $y = 2 \sin 3x$ for $-\pi \leq x \leq \pi$, using at least $\frac{1}{3}$ of the page. 2

(c) Prove the identity: 3

$$\frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} + \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = 2 \sec 2\theta$$

(d) The base camp (B) of an expedition is located x metres due south of the base of a mountain (M) with height h metres. Amelie (A) moved 30 metres due west of the camp and spots a ruin (R) at the top of the mountain peak with her scope showing an angle of elevation of 45° . At the same time, Clement (C) was 90 metres directly south-east of the camp and spots the same ruin with his scope showing an angle of elevation of 30°



(i) Show that $h^2 = x^2 + 900$. 1

(ii) Show that $MC^2 = x^2 + 90\sqrt{2}x + 90^2$ 2

(iii) Find the value of h to the nearest metre. 3

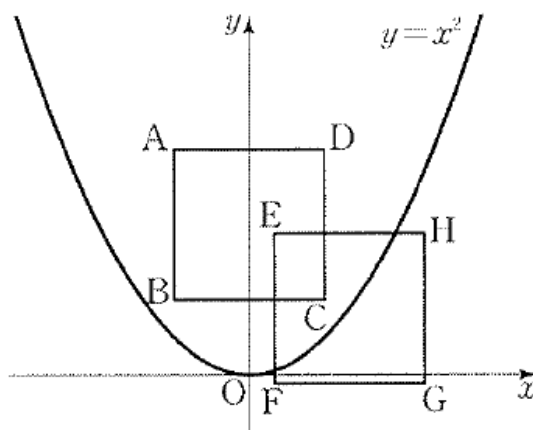
Question 10 (18 marks)

- (a) The line $3x - 4y = 4$ is more than 2 units from the point $(-2, \delta)$. Find the possible values of δ . 2
- (b) Give a geometric description of the locus of a point P which moves so that $PA=2PB$, where $P = (x, y)$, $A = (-2, 4)$ and $B = (4, 1)$. 3
- (c) $P(2ap, -ap^2)$ and $Q(2aq, -aq^2)$ are two points on the parabola $x^2 = -4ay$.
- (i) Find the equation of the chord of PQ . 2
- (ii) What is the value of pq when PQ is a focal chord? 2
- (d) P and Q are two points on the parabola $x^2 = 4ay$ with coordinates $(2ap, ap^2)$ and $(2aq, aq^2)$ respectively. The tangents at P and Q meet at T , which is situated on the parabola $x^2 = -4ay$.
- (i) Write down the equations of the tangents at P and Q . 2
- (ii) Show that T is the point $(a(p + q), apq)$. 2
- (iii) Prove that $p^2 + q^2 = -6pq$. 3
- (iv) Find the equation of the locus of the midpoint of PQ . 2

Question 11 (9 marks)

(a) Find the function $f(x)$ that satisfies $2f(x) + 3f(1-x) = x^2$ for all real x . 3

(b) From the following diagram, the square $ABCD$ has a side length of 1 and its diagonals intersect at $(0,1)$. The square $EFGH$ also has a side length of 1 and its diagonals intersect on the parabola $y = x^2$. (All sides of the squares are parallel to the x and y axes)



(i) Show that the coordinates of C are $(\frac{1}{2}, \frac{1}{2})$ 1

(ii) Let the intersection of diagonals in $EFGH$ be (p, p^2) and find the coordinates of E in terms of p . 1

(iii) Show that the area equation of the overlapping region of square $ABCD$ and $EFGH$ is 2

$$A(p) = -p^3 + p^2$$

(iv) What is the maximum area of the overlapping region? 2

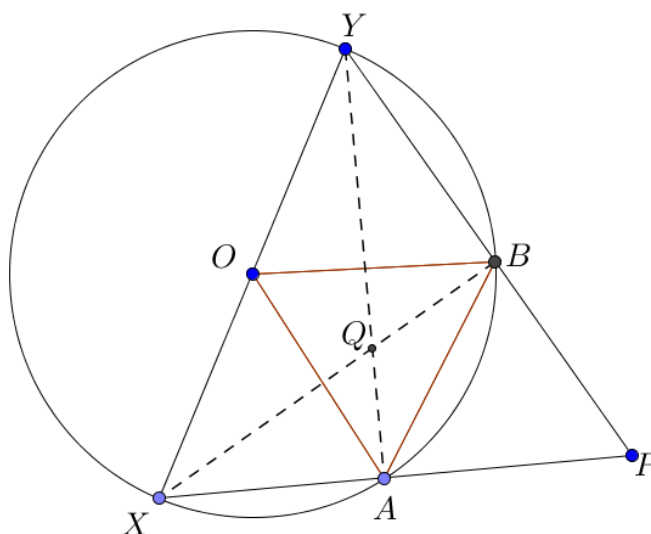
Question 12 (10 marks)

- (a) Suppose that $\sin \alpha + \sin \beta = \sqrt{\frac{5}{3}}$ and $\cos \alpha + \cos \beta = 1$.

Find the value of $\cos(\alpha - \beta)$.

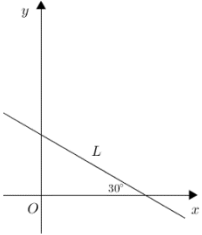
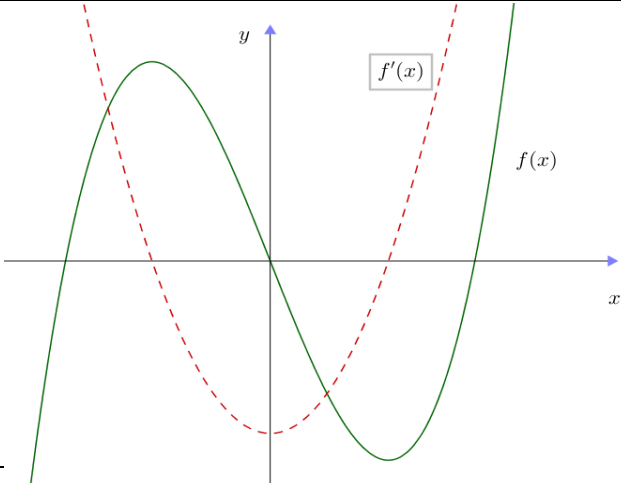
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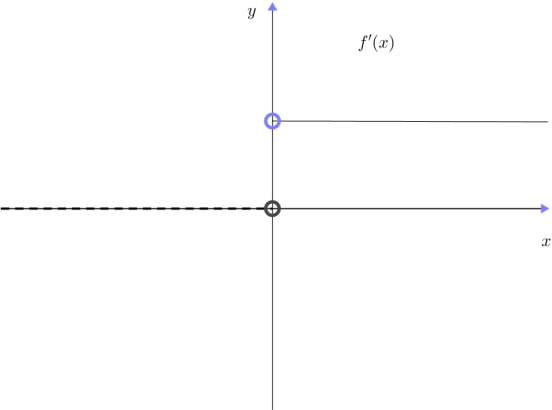
- (b) In a circle centred at O , points A and B are chosen so that OAB forms an equilateral triangle. Let points X and Y be two points of a diameter, and the intersection between lines XA and YB produced be P . Connect points AY and BX and let the intersection of AY and BX be Q .

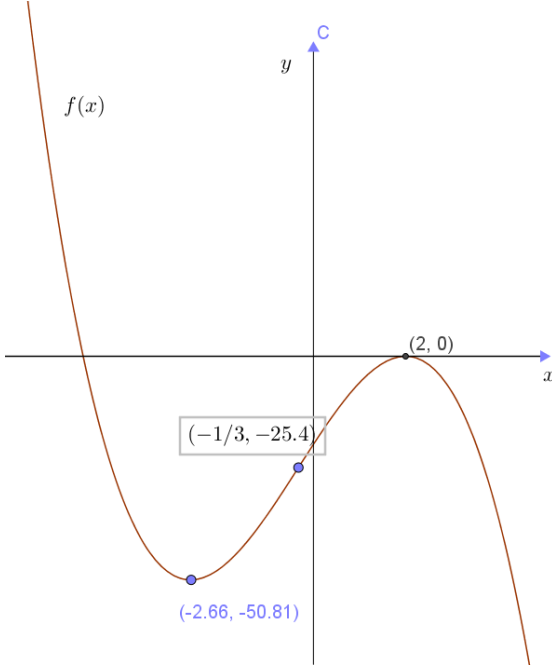


- | | |
|--|----------|
| (i) Show that $\angle PAQ = 90^\circ$. | 1 |
| (ii) Show that $APBQ$ is a cyclic quadrilateral. | 2 |
| (iii) State why $\angle XQA = \angle XYA + \angle YXB$. | 1 |
| (iv) Hence, find the size of $\angle APB$. | 2 |
| (v) Describe the locus of P as X and Y move around the circle. | 1 |

End of paper

Section 1 - Multiple Choice	
1	$y = \frac{x^2}{x^2 - 4x - 12} = \frac{x^2}{(x - 6)(x + 2)}$ $\therefore$ 2 vertical asymptotes + 1 horizontal asymptote = 3 (C)
2	$(x + 3)^2 = 12(y - 3)$ $V = (-3, 3), a = 3$ $\therefore S = (-3, 6)$ (C)
3	$\frac{\text{rise}}{\text{run}} = -\frac{1}{\sqrt{3}}$ (B) 
4	$\tan \alpha = -\frac{3}{4}, \tan \beta = -\frac{4}{3}$ $\tan(\alpha - \beta) = \frac{-\frac{3}{4} + \frac{4}{3}}{1 + 1} = \frac{7}{24}$ (D)
5	$f(x) \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ 2x, & x < 0 \end{cases}, \quad f^{-1}(x) \begin{cases} x, & x > 0 \\ 0, & x = 0 \\ \frac{x}{2}, & x < 0 \end{cases}$ $a \checkmark, b \checkmark, c \times$ (A)
Section 2	
6a) (i)	

(ii)	 $x = 0, \text{ when } x < 0$
b)	$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $\lim_{h \rightarrow 0} \frac{(3(x+h)^2 - 7(x+h)) - 3x^2 + 7x}{h} = \lim_{h \rightarrow 0} \frac{3(2x+h)h - 7h}{h}$ $= \lim_{h \rightarrow 0} 3(2x+h) - 7 = 6x - 7$
c) (i)	$y = 4x^2 + 3x^{-\frac{1}{2}}$ $y' = 8x - \frac{3}{2}x^{-\frac{3}{2}} = 8x - \frac{3}{2\sqrt{x^3}}$
(ii)	$y = (1-2x)^5(2x+1)$ $y' = -10(1-2x)^4(2x+1) + 2(1-2x)^5$ $= (1-2x)^4\{2(1-2x) - 10(2x+1)\}$ $= (1-2x)^4(-8-24x)$ $= -8(1-2x)^4(1+3x)$
(iii)	$y = \frac{5x^2 + 3}{1-x}$ $y' = \frac{10x(1-x) + 5x^2 + 3}{(1-x)^2} = \frac{3 + 10x - 5x^2}{(1-x)^2}$

7 a)	$y = 2x^3 - 10x + 1, \quad x = 2 \therefore y = 16 - 20 + 1 = -3$ $y' = 6x^2 - 10, \quad \text{at } x = 2, \quad y' = 24 - 10 = 14$ $m_{normal} = -\frac{1}{14}$ $y + 3 = -\frac{1}{14}(x - 2)$ $y = -\frac{x}{14} - \frac{20}{7}$								
b) (i)	$y = -x^3 - x^2 + 16x - 20$ $y' = -3x^2 - 2x + 16 = -(3x + 8)(x - 2)$ $y' = 0 \text{ at } x = 2 \text{ or } x = -\frac{8}{3}$ $y'' = -6x - 2$ $\text{at } x = 2, y'' = -14 < 0 \text{ local max}$ $\text{at } x = -\frac{8}{3}, y'' = 14 > 0 \text{ local min}$ $(2,0) \text{ local max}, \left(-\frac{8}{3}, -50.8\right) \text{ local min}$								
(ii)	$y'' = -6x - 2 = 0$ $x = -\frac{1}{3}, y = -25.4$ <table><tr><td>x</td><td>-1</td><td>$-\frac{1}{3}$</td><td>0</td></tr><tr><td>y''</td><td>4</td><td>0</td><td>-2</td></tr></table> $\therefore \left(-\frac{1}{3}, -25.4\right) \text{ is a point of inflexion}$	x	-1	$-\frac{1}{3}$	0	y''	4	0	-2
x	-1	$-\frac{1}{3}$	0						
y''	4	0	-2						
(iii)									

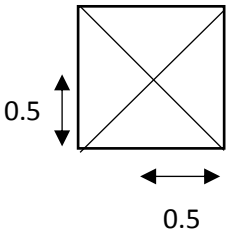
8 a)	$(i) \alpha + \beta = -\frac{b}{a} = \frac{4}{3}$ $(ii) \alpha\beta = \frac{c}{a} = \frac{2}{3}$ $(iii) \frac{2}{\alpha} + \frac{2}{\beta} = \frac{2(\alpha + \beta)}{\alpha\beta} = \frac{2\left(\frac{4}{3}\right)}{\frac{2}{3}} = 4$ $(iv) (\alpha - \beta)(\beta - \alpha) = -(\alpha - \beta)^2$ $= -\{(\alpha + \beta)^2 - 4\alpha\beta\}$ $= 4\alpha\beta - (\alpha + \beta)^2$ $= 4\left(\frac{2}{3}\right) - \left(\frac{4}{3}\right)^2 = \frac{8}{3} - \frac{16}{9}$ $= \frac{8}{9}$
b)	$3x^2 - 2x + 5 \equiv Ax^2 - 2Ax + Bx - 2B + C$ $A = 3 \text{ (only coefficient of } x^2)$ $B - 2A = -2, \therefore B = -2 + 2A = 4$ $C - 2B = 5, \quad \therefore C = 5 + 2B = 13$
c)	$\text{let } 3^x = t$ $t^2 - 12t + 27 = 0$ $(t - 3)(t - 9) = 0, \therefore t = 3, 9$ $3^x = 3, \therefore x = 1$ $3^x = 9, \therefore x = 2$
d) (i)	$\text{real, } \Delta \geq 0$ $\Delta = (4 - m)^2 - 4(2m - 3)$ $= 16 - 8m + m^2 - 8m + 12$ $= m^2 - 16m + 28 \geq 0$ $(m - 14)(m - 2) \geq 0$ $\therefore m \leq 2 \text{ or } m \geq 14$
(ii)	$\Delta \geq 0,$ $\text{and } m - 4 < 0, \therefore m < 4$ $\text{and } 2m - 3 > 0, \therefore m > \frac{3}{2}$ $\frac{3}{2} < m < 4$ $\text{(Draw a number line)}$

9a)	$\operatorname{cosec} 240^\circ = \frac{1}{\sin 240^\circ} = -\frac{1}{\sin 60} = -\frac{1}{\frac{\sqrt{3}}{2}} = -\frac{2}{\sqrt{3}} \text{ or } -\frac{2\sqrt{3}}{3}$
b)	
c)	$\begin{aligned} LHS &= \frac{\cos \theta + \sin \theta}{(\cos \theta - \sin \theta)} + \frac{\cos \theta - \sin \theta}{(\cos \theta + \sin \theta)} \\ &= \frac{(\cos \theta + \sin \theta)^2 + (\cos \theta - \sin \theta)^2}{(\cos \theta - \sin \theta)(\cos \theta + \sin \theta)} \\ &= \frac{\cos^2 \theta + 2 \sin \theta \cos \theta + \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta + \sin^2 \theta}{\cos^2 \theta - \sin^2 \theta} \\ &= \frac{2(\cos^2 \theta + \sin^2 \theta)}{\cos^2 \theta - \sin^2 \theta}, (\text{since } \cos^2 \theta + \sin^2 \theta = 1, \cos^2 \theta - \sin^2 \theta = \cos 2\theta) \\ &= \frac{2}{\cos 2\theta} = 2 \sec 2\theta = RHS \end{aligned}$
d) (i)	ΔRMA is a right angled at M, $AM = h$, $\left(\tan 45^\circ = \frac{h}{AM} = 1 \right)$ $AM^2 = 30^2 + x^2 \therefore h^2 = 30^2 + x^2$
(ii)	From cosine rule $MC^2 = x^2 + 90^2 - 2 \times 90 \times x \times \cos \angle MBC, \text{ Where } \angle MBC = 90 + 45 = 135^\circ$ $= x^2 + 90^2 - 2 \times 90 \times x \times -\frac{1}{\sqrt{2}} = x^2 + 90^2 + 90\sqrt{2}x$ <i>Q.E.D.</i>
(iii)	From (i) and (ii) From RMC, $\frac{h^2}{MC^2} = \frac{x^2 + 900}{x^2 + 90\sqrt{2} + 90^2} = \tan^2 30^\circ = \frac{1}{3}$ $3x^2 + 2700 = x^2 + 90\sqrt{2} + 90^2$ $2x^2 - 90\sqrt{2}x - 5400 = 0$ $x^2 - 45\sqrt{2}x - 2700 = 0$ $\therefore x = \frac{45\sqrt{2} + \sqrt{4050 - 4(1)(-2700)}}{2}, \text{ as } x > 0.$ $h = \sqrt{x^2 + 900} = 97m \text{ (nearest } m)$

10 a)	$\text{perpendicular distance} = \frac{ 3 \times (-2) - 4 \times \delta - 4 }{\sqrt{3^2 + 4^2}} = \frac{ -10 - 4\delta }{5}$ $\frac{ -10 - 4\delta }{5} > 2, \quad \therefore -10 - 4\delta > 10$ $(-10 - 4\delta)^2 > 100$ $4(5 + 2\delta)^2 > 100, \quad \therefore (5 + 2\delta)^2 > 25$ $\delta > 0 \text{ or } \delta < -5$
b)	$PA = 2PB, \therefore PA^2 = 4PB^2$ $(x + 2)^2 + (y - 4)^2 = 4\{(x - 4)^2 + (y - 1)^2\}$ $x^2 + 4x + 4 + y^2 - 8y + 16 = 4(x^2 - 8x + 16 + y^2 - 2y + 1)$ $4x^2 - 32x + 72 + 4y^2 - 8y + 4 - x^2 - 4x - 4 - y^2 + 8y - 16 = 0$ $3x^2 - 36x + 3y^2 + 56 = 0$ $x^2 - 12x + y^2 = -16$ $(x - 6)^2 + y^2 = 20$ It is a circle locus with centre : (6,0) with radius of $2\sqrt{5}$
c) (i)	$m_{PQ} = \frac{-aq^2 + ap^2}{2aq - 2ap} = \frac{a(p - q)(p + q)}{-2a(p - q)} = -\frac{p + q}{2}$ $y + ap^2 = -\frac{p + q}{2}(x - 2ap)$ $y = -\frac{(p + q)}{2}x + ap(p + q) - ap^2$ $y = -\frac{p + q}{2}x + apq$
(ii)	PQ as focal chord passes through (0,-a) Sub (0,-a) into the equation from (i) $-a = apq$ $\therefore pq = -1$
d) (i)	$y - ap^2 = p(x - 2ap)$ $y = px - ap^2$ Similarly $y = qx - aq^2$
(ii)	$px - ap^2 = qx - aq^2$ $(p - q)x = a(p - q)(p + q)$ $x_T = a(p + q)$ $y_T = p\{a(p + q)\} - ap^2 = apq, \quad \therefore T(a(p + q), apq)$
(iii)	T lies on $x^2 = -4ay$ $a^2(p + q)^2 = -4a(apq)$ $a^2[p^2 + q^2 + 2pq] = -4a^2pq$ $p^2 + q^2 = -6pq$

(iv)

$$\begin{aligned} & M\left(a(p+q), \frac{a(p^2+q^2)}{2}\right) \\ & x = a(p+q), \quad \frac{2y}{a} = p^2+q^2 \\ & x^2 = a^2(p+q)^2 = a^2(p^2+q^2+2pq) \\ & \quad = \frac{1}{3}a^2(3(p^2+q^2)+6pq) \\ & \quad = \frac{2}{3}a^2(p^2+q^2) \\ & \quad = \frac{2}{3}a^2\left(\frac{2y}{a}\right) = \frac{4}{3}ay \\ & \therefore x^2 = \frac{4}{3}ay \end{aligned}$$

11 a)	$2f(x) + 3f(1-x) = x^2 \quad - (1)$ Let $x = 1 - x$ $2f(1-x) + 3f(x) = (1-x)^2 \quad - (2)$ $3 \times (2) - 2 \times (1)$ $6f(1-x) + 9f(x) - 4f(x) - 6f(1-x) = 3x^2 - 6x + 3 - 2x^2$ $5f(x) = x^2 - 6x + 3$ $\therefore f(x) = \frac{1}{5}(x^2 - 6x + 3)$
b) (i)	 $C = \left(0 + \frac{1}{2}, 1 - \frac{1}{2}\right)$ $= \left(\frac{1}{2}, \frac{1}{2}\right)$
(ii)	Like part (i) $E = \left(p - \frac{1}{2}, p^2 + \frac{1}{2}\right)$
(iii)	$A(p) = b \times h = \left(\frac{1}{2} - \left(p - \frac{1}{2}\right)\right) \left(\left(p^2 + \frac{1}{2}\right) - \frac{1}{2}\right)$ $A(p) = (1-p)p^2 = -p^3 + p^2$
(iv)	$\frac{dA}{dp} = -3p^2 + 2p = p(-3p + 2)$ Stationary point = $p(-3p + 2) = 0$ $p = 0 \text{ or } p = \frac{2}{3}$ $\frac{d^2A}{dp^2} = -6p + 2$ $f''(0) = 2, \quad \text{Min}$ $f''\left(\frac{2}{3}\right) = -2, \quad \text{Max}$ $\therefore \text{Max } A = -\left(\frac{2}{3}\right)^3 + \left(\frac{2}{3}\right)^2 = -\frac{8}{27} + \frac{12}{27} = \frac{4}{27} u^2$

12 a)	$\sin \alpha + \sin \beta = \sqrt{\frac{5}{3}} \quad (1)$ $\cos \alpha + \cos \beta = 1 \quad (2)$ $(1)^2 + (2)^2 = \sin^2 \alpha + 2 \sin \alpha \sin \beta + \sin^2 \beta + \cos^2 \alpha + 2 \cos \alpha \cos \beta + \cos^2 \beta$ $= \sin^2 \alpha + \cos^2 \alpha + \sin^2 \beta + \cos^2 \beta + 2 \cos \alpha \cos \beta + 2 \sin \alpha \sin \beta$ $= 2 + 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta) = 2 + 2 \cos(\alpha - \beta)$ $\frac{5}{3} + 1 = 2 + 2 \cos(\alpha - \beta)$ $\therefore \cos(\alpha - \beta) = \frac{1}{3}$
b) (i)	$\angle PAQ = 180 - \angle XAY$ (Supplementary angle) $\angle XAY = 90^\circ$ (angle in a semicircle) $\therefore \angle PAQ = 90^\circ$ as given
(ii)	Similarly $\angle PBQ = 90^\circ$ $\angle PBQ + \angle PAQ = 90^\circ + 90^\circ = 180^\circ$ opposite angles are supplementary $\therefore$ APBQ is a cyclic quadrilateral
(iii)	$\angle XQA = \angle XYA + \angle YXB,$ (exterior angle of a triangle is equal to the sum of other two interior angles)
(iv)	$\angle XYA = \frac{1}{2} \angle XOA, \angle YXB = \frac{1}{2} \angle YOB$ $\angle XOA + \angle YOB = 180^\circ - 60^\circ$ (equilateral Δ) $= 120^\circ$ $\therefore \angle XYA + \angle YXB = \angle XQA = 60^\circ$ $\angle APB = \angle XQA = 60^\circ$ (exterior angle of cyclic quadrilateral is equal to the opposite interior angle)
(v)	Point P traces circumference of a circle subtended from arc AB with angle at the circumference of 60° with diameter of PQ.